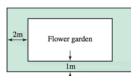


Problem 2

Problem 2

Problem 1 A rectangular flower garden with an area of 30 m^2 is surrounded by a grass border 1 m wide on two sides and 2 m wide on the other two sides (see figure). What dimensions of the garden minimize the combined area of the garden and borders?



- Label the picture with variables.
- What are you trying to maximize or minimize? Write an equation for it in terms of the variables from (a).
- What is your constraint? Write a constraint equation in terms of the variables from (a).
- Reduce your optimization equation to one variable using the constraint equation.

Problem 2

- (e) *What is the interval on which your variable makes sense? Is it open or closed? What does this mean for the method of finding the global max or min?*
- (f) *Use the appropriate method to find and justify your global extremum.*
- (g) *Be sure to answer the question asked in the original problem.*